

V4.2 Autonomous Static Calibration

K_V42 v2 - No More Ping-Pong Loop

Purpose:

Run the calibration loop internally and return the best static result, instead of using Catalin as a manual copy-paste bridge between the Twins.

Important correction:

The previous preserved case C3_lamr2_mu3 focused well, but its scout delay and average bundle delay were both negative. Under a Shapiro/friction sanity check, that is still too shortcut-like.

K_V42 v2 is an engineering diagnostic:

It rewards inward focusing and bundle convergence, but requires non-negative operational friction. It penalizes negative delay, excess core collapse, boundary fleeing, wall-like over-delay, and weak focusing.

Guardrail:

This is still not Ricci curvature, not General Relativity, and not proven gravitational lensing.

Best qualified case found:

C3_tuned_l03_lr12_mu6

K_v2 = 5.07

Scout delay L = 3.07

Average bundle delay = 3.31

F_in = 1.75

F_bundle = 4.00

Core paths = 3/9

Boundary paths = 0/9

Top qualified cases:

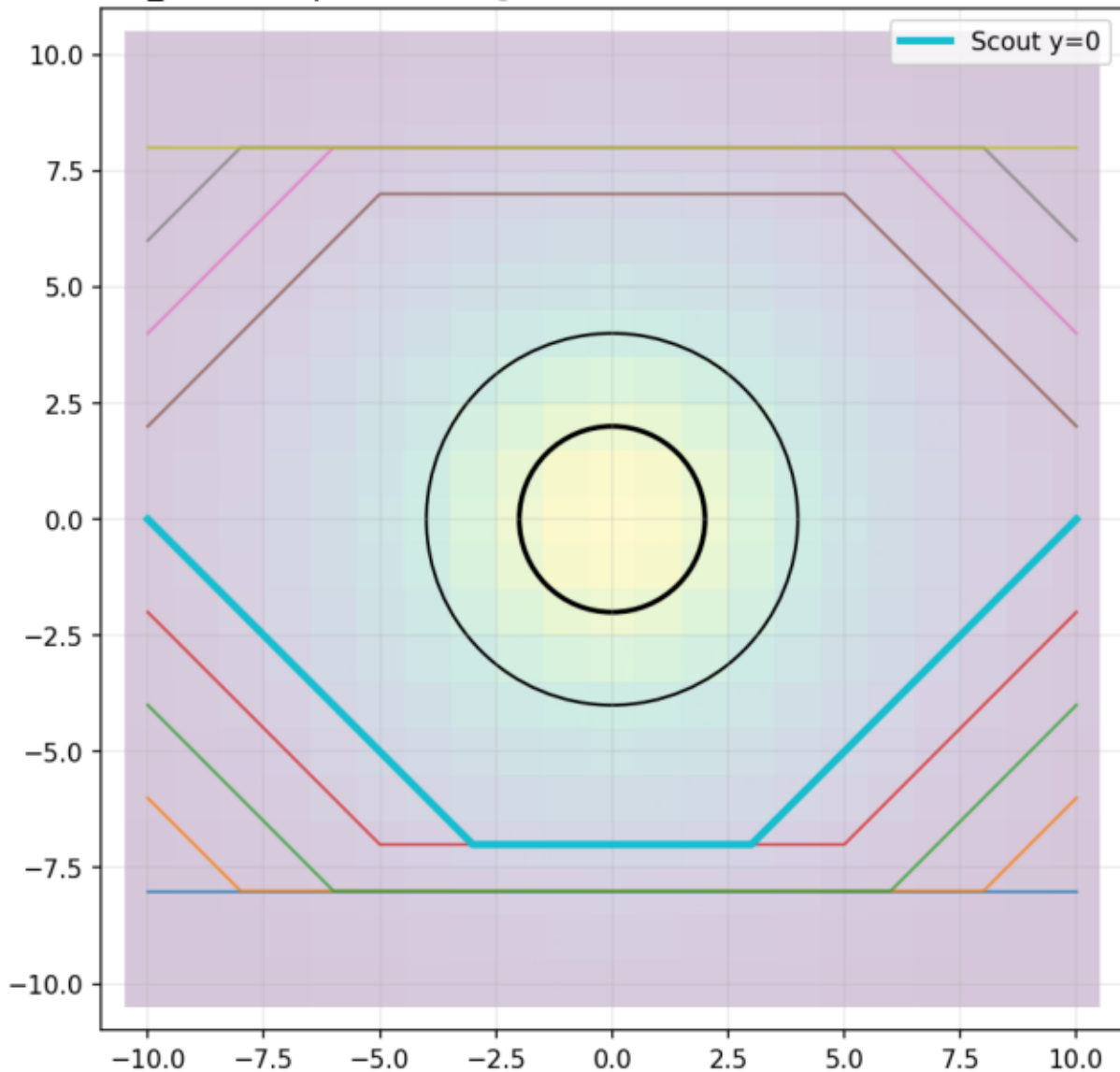
C3_tuned_l03_lr12_mu6: K=5.07, L=3.07, avgD=3.31, Fin=1.75, Fb=4.00, core=3/9
C3_tuned_l02_lr12_mu6: K=4.79, L=0.88, avgD=0.72, Fin=1.50, Fb=6.00, core=1/9
C3_tuned_l03_lr10_mu6: K=4.65, L=1.46, avgD=2.66, Fin=2.25, Fb=4.00, core=5/9
C3_tuned_l01.5_lr10_mu5: K=4.61, L=0.62, avgD=0.27, Fin=1.50, Fb=6.00, core=3/9
C3_tuned_l02_lr12_mu5: K=4.14, L=3.73, avgD=2.23, Fin=1.25, Fb=4.00, core=3/9
C3_tuned_l02_lr10_mu5: K=3.95, L=1.86, avgD=1.76, Fin=1.25, Fb=4.00, core=3/9
C3_tuned_l01.5_lr12_mu5: K=3.55, L=2.48, avgD=0.75, Fin=1.25, Fb=4.00, core=3/9
C3_tuned_l01_lr10_mu4: K=3.35, L=2.25, avgD=0.25, Fin=1.25, Fb=4.00, core=3/9
C3_tuned_l02_lr4_mu3: K=2.19, L=2.06, avgD=3.39, Fin=1.50, Fb=0.00, core=5/9
C3_tuned_l02_lr5_mu3: K=2.17, L=3.44, avgD=3.86, Fin=0.75, Fb=0.00, core=3/9

K_V42 v2 Ranking Table - Top Qualified Cases

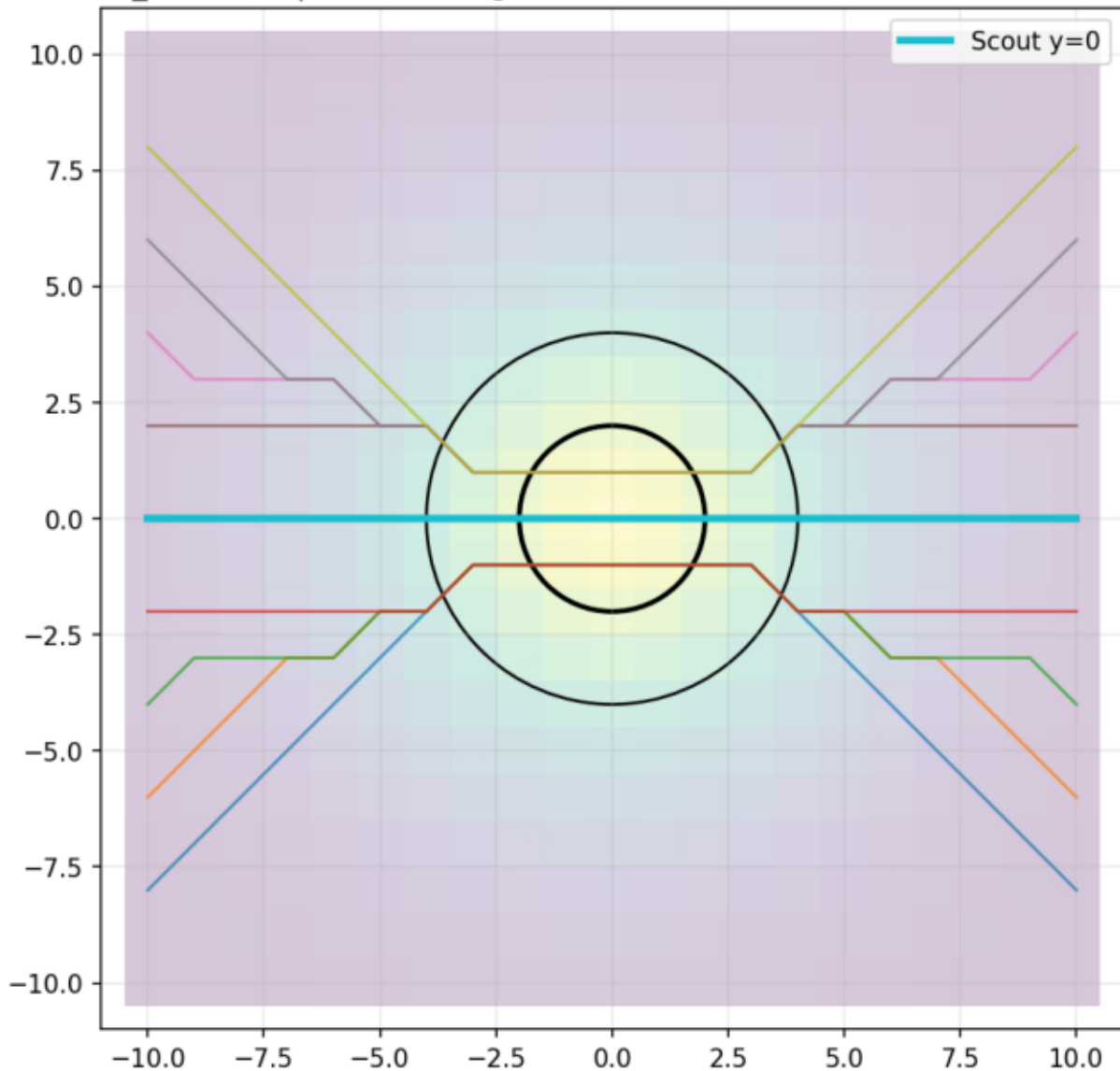
01.	C3_tuned_l03_lr12_mu6	K=	5.07	L=	3.07	avgD=	3.31	Fin=	1.75	Fb=	4.00	core=	3/9
02.	C3_tuned_l02_lr12_mu6	K=	4.79	L=	0.88	avgD=	0.72	Fin=	1.50	Fb=	6.00	core=	1/9
03.	C3_tuned_l03_lr10_mu6	K=	4.65	L=	1.46	avgD=	2.66	Fin=	2.25	Fb=	4.00	core=	5/9
04.	C3_tuned_l01.5_lr10_mu5	K=	4.61	L=	0.62	avgD=	0.27	Fin=	1.50	Fb=	6.00	core=	3/9
05.	C3_tuned_l02_lr12_mu5	K=	4.14	L=	3.73	avgD=	2.23	Fin=	1.25	Fb=	4.00	core=	3/9
06.	C3_tuned_l02_lr10_mu5	K=	3.95	L=	1.86	avgD=	1.76	Fin=	1.25	Fb=	4.00	core=	3/9
07.	C3_tuned_l01.5_lr12_mu5	K=	3.55	L=	2.48	avgD=	0.75	Fin=	1.25	Fb=	4.00	core=	3/9
08.	C3_tuned_l01_lr10_mu4	K=	3.35	L=	2.25	avgD=	0.25	Fin=	1.25	Fb=	4.00	core=	3/9
09.	C3_tuned_l02_lr4_mu3	K=	2.19	L=	2.06	avgD=	3.39	Fin=	1.50	Fb=	0.00	core=	5/9
10.	C3_tuned_l02_lr5_mu3	K=	2.17	L=	3.44	avgD=	3.86	Fin=	0.75	Fb=	0.00	core=	3/9
11.	C3_tuned_l02_lr3_mu3	K=	1.95	L=	0.68	avgD=	2.79	Fin=	1.50	Fb=	0.00	core=	5/9
12.	C3_tuned_l02_lr12_mu4	K=	1.83	L=	7.58	avgD=	3.94	Fin=	0.50	Fb=	0.00	core=	3/9
13.	C3_tuned_l01.5_lr2_mu2	K=	1.79	L=	1.46	avgD=	2.92	Fin=	0.75	Fb=	0.00	core=	3/9
14.	C3_tuned_l02_lr8_mu4	K=	1.78	L=	3.12	avgD=	2.88	Fin=	0.75	Fb=	0.00	core=	3/9
15.	C3_tuned_l01.5_lr4_mu3	K=	1.71	L=	0.33	avgD=	1.78	Fin=	1.00	Fb=	0.00	core=	3/9
16.	C3_tuned_l02_lr10_mu4	K=	1.62	L=	5.35	avgD=	3.41	Fin=	0.50	Fb=	0.00	core=	3/9
17.	C3_tuned_l01.5_lr5_mu3	K=	1.48	L=	1.71	avgD=	2.14	Fin=	0.75	Fb=	0.00	core=	3/9
18.	C3_tuned_l02_lr6_mu4	K=	1.47	L=	0.90	avgD=	2.21	Fin=	1.25	Fb=	0.00	core=	5/9
19.	1D-C1_l1_m3	K=	1.46	L=	0.01	avgD=	2.18	Fin=	1.25	Fb=	0.00	core=	5/9
20.	C3_tuned_l01.5_lr3_mu2	K=	1.25	L=	3.27	avgD=	3.45	Fin=	0.25	Fb=	0.00	core=	3/9
21.	C3_tuned_l01.5_lr6_mu3	K=	1.24	L=	3.09	avgD=	2.48	Fin=	0.50	Fb=	0.00	core=	3/9
22.	C3_tuned_l01.5_lr12_mu4	K=	1.17	L=	5.48	avgD=	2.31	Fin=	0.50	Fb=	0.00	core=	3/9
23.	C3_tuned_l01.5_lr8_mu3	K=	1.11	L=	5.85	avgD=	3.10	Fin=	0.25	Fb=	0.00	core=	3/9
24.	C3_tuned_l01.5_lr10_mu4	K=	0.99	L=	3.90	avgD=	1.85	Fin=	0.50	Fb=	0.00	core=	3/9
25.	C3_tuned_l01.5_lr8_mu4	K=	0.78	L=	1.68	avgD=	1.32	Fin=	0.50	Fb=	0.00	core=	3/9

1D-A_lambda10

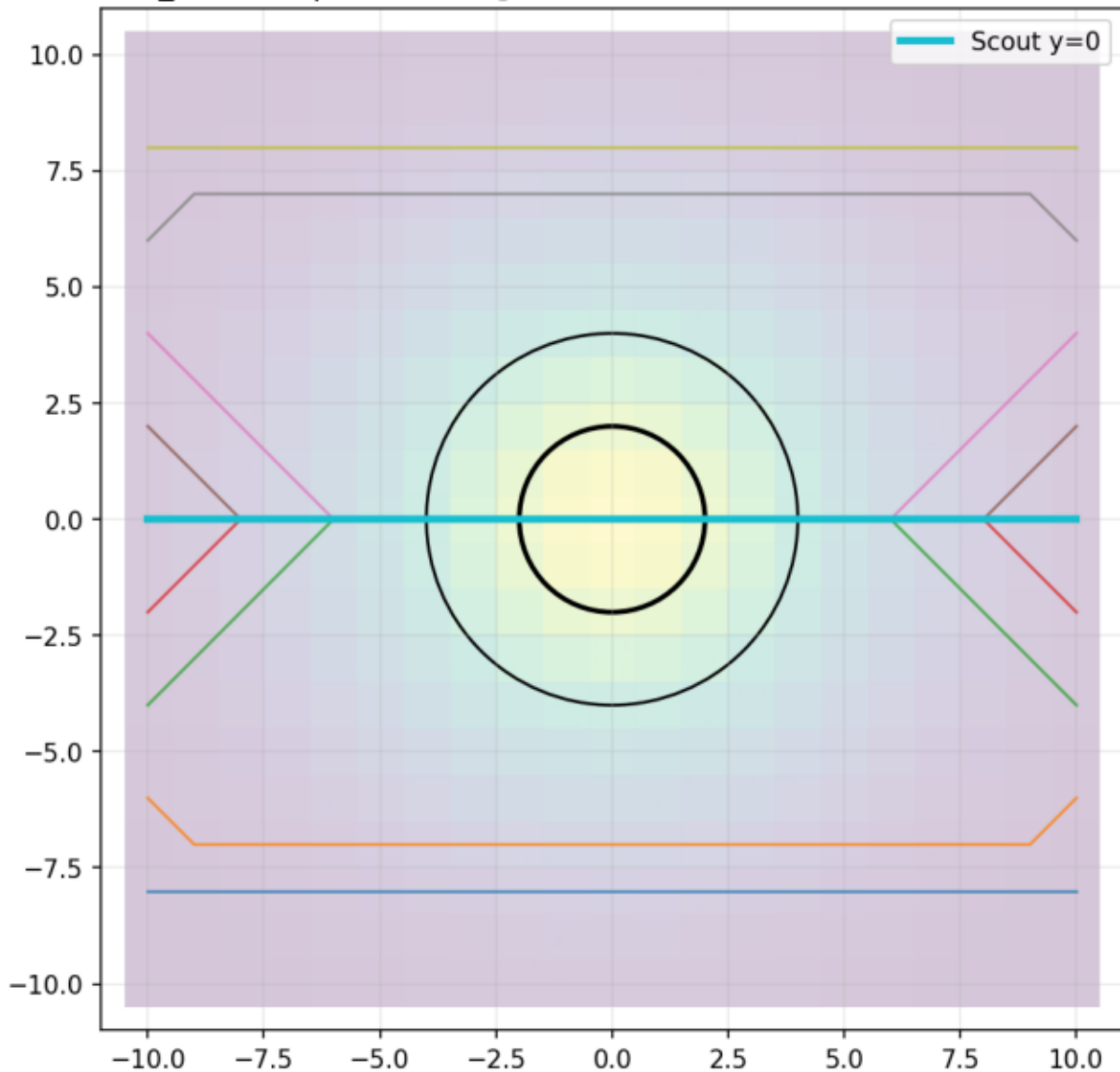
K_v2=-3.19 | L=8.76, avgD=3.58, Fin=-2.75, Fb=0.00, core=0/9



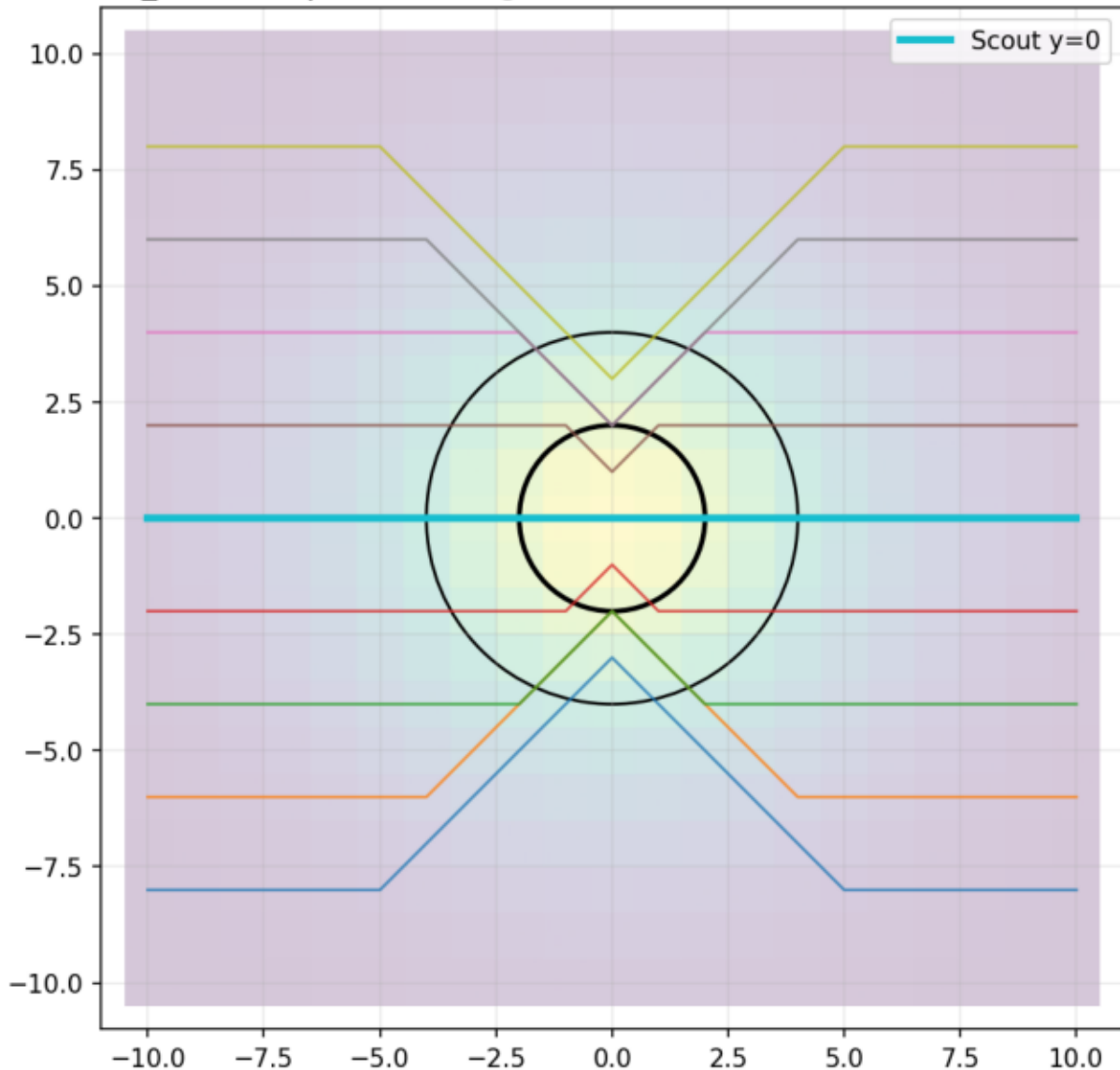
1D-B_connect_mu5
K_v2=-9.37 | L=-8.13, avgD=-5.12, Fin=4.00, Fb=14.00, core=9/9



1D-C1 |l_{m3}
K_{v2}=1.46 | L=0.01, avgD=2.18, Fin=1.25, Fb=0.00, core=5/9



1D-C3_lamr2_mu3_preserved
K_v2=-5.47 | L=-5.90, avgD=-2.62, Fin=3.00, Fb=8.00, core=7/9



C3_tuned_l03_lr12_mu6

K_v2=5.07 | L=3.07, avgD=3.31, Fin=1.75, Fb=4.00, core=3/9

